TECHBIZ ACADEMY

INTERNSHIP TASK REPORT

WEEK 1 — TASK 2

IPv4 Addressing and Subnetting

DETAIL	INFORMATION
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Task Number	Week 1 – Task 2
Task Title	IPv4 Addressing and Subnetting
Maximum Score	10 Quality Points
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Prepared for practical networking internship assessment

1. Introduction and Objectives

This report covers IPv4 addressing, binary conversion, subnet masks, network and broadcast addresses, usable host ranges, and Variable Length Subnet Masking (VLSM). The practical objective is to divide the network 192.168.10.0/24 into five appropriately sized subnets and then apply the addressing plan to the Week 1 Task 1 Packet Tracer lab.

The VLSM plan and calculations in this report are a worked addressing plan. The Week 1 Task 1 topology-specific interface assignments and connectivity results must be completed using the student's own lab and verified screenshots before submission.

2. IPv4 Binary Conversion

An IPv4 address contains 32 bits divided into four 8-bit octets. Each bit position has a decimal weight: 128, 64, 32, 16, 8, 4, 2 and 1. To convert decimal to binary, select the bit weights that add up to the decimal value. To convert binary back to decimal, add the weights whose bits are 1.

Decimal	8-bit binary	Working / reverse check
0	00000000	0 = 0
1	00000001	1 = 1
10	00001010	8 + 2 = 10
25	00011001	16 + 8 + 1 = 25
50	00110010	32 + 16 + 2 = 50
64	01000000	64 = 64
100	01100100	64 + 32 + 4 = 100
128	10000000	128 = 128
192	11000000	128 + 64 = 192
255	11111111	128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

Verification: each binary value contains exactly eight bits. The reverse-check column confirms that converting the binary value back to decimal produces the original octet.

3. Important IPv4 and Subnetting Terms

Term	Explanation
IPv4 address	A 32-bit logical address used to identify a device/interface on an IP network.
Subnet mask	A 32-bit value that identifies the network portion and host portion of an IPv4 address.
Prefix length	The number of network bits. For example, /24 means 24 network bits and 8 host bits.
Network address	The first address in a subnet; it identifies the subnet and is not assigned to a normal host.
Broadcast address	The last address in a subnet; it is used to send traffic to all hosts in that subnet.
Usable host range	The assignable host addresses between the network and broadcast addresses in a traditional subnet.
Default gateway	The router interface address a host uses to reach destinations outside its local subnet.

For a conventional IPv4 subnet, usable hosts = 2^(host bits) − 2. The two reserved addresses are the network address and broadcast address. A /30 has four total addresses and two usable host addresses.

4. VLSM Design for 192.168.10.0/24

The subnet requirements are 50, 25, 12, 2 and 2 hosts. VLSM is applied from the largest host requirement to the smallest so that address space is used efficiently.

Required hosts	Chosen prefix	Subnet mask	Total addresses	Usable hosts
50	/26	255.255.255.192	64	62
25	/27	255.255.255.224	32	30

12	/28	255.255.255.240	16	14
2	/30	255.255.255.252	4	2
2	/30	255.255.255.252	4	2

Subnet	Network address / prefix	Subnet mask	Usable host range	Broadcast
A — 50 hosts	192.168.10.0/26	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63
B — 25 hosts	192.168.10.64/27	255.255.255.224	192.168.10.65 – 192.168.10.94	192.168.10.95
C — 12 hosts	192.168.10.96/28	255.255.255.240	192.168.10.97 – 192.168.10.110	192.168.10.111
D — 2 hosts	192.168.10.112/30	255.255.255.252	192.168.10.113 – 192.168.10.114	192.168.10.115
E — 2 hosts	192.168.10.116/30	255.255.255.252	192.168.10.117 – 192.168.10.118	192.168.10.119

Address allocation check: the five subnets occupy 192.168.10.0 through 192.168.10.119. There is no overlap, and the remaining range 192.168.10.120–192.168.10.255 is still available for future subnetting.

5. Subnetting Method and Mental Shortcuts

- Sort the host requirements from largest to smallest before allocating subnets.
- Choose the smallest block that can hold the required hosts plus network and broadcast addresses.
- Block size = 256 − the interesting subnet-mask octet. For /26, 256 − 192 = 64; network boundaries are 0, 64, 128 and 192.
- For /27, the block size is 32; for /28 it is 16; for /30 it is 4.
- The network address is the start of the block. The broadcast address is one less than the next block boundary.
- The usable range starts one address after the network address and ends one address before the broadcast address.

6. Re-addressing the Week 1 Task 1 Lab

Use the VLSM subnets above to re-address the existing Week 1 Task 1 topology. Select the subnet for each LAN or point-to-point link according to its actual host requirement. Assign a unique usable IP address to every router interface and PC, use the matching subnet mask, and configure the appropriate default gateway on each PC.

Device / Interface	Assigned subnet	IP address	Subnet mask	Default gateway / notes
Router1 – LAN	Subnet A (/26)	192.168.10.1	255.255.255.192	Default gateway for PC1, PC2
PC1	Subnet A (/26)	192.168.10.2	255.255.255.192	192.168.10.1
PC2	Subnet A (/26)	192.168.10.3	255.255.255.192	192.168.10.1
Router2 – LAN	Subnet B (/27)	192.168.10.65	255.255.255.224	Default gateway for PC3, PC4
PC3	Subnet B (/27)	192.168.10.66	255.255.255.224	192.168.10.65
PC4	Subnet B (/27)	192.168.10.67	255.255.255.224	192.168.10.65
Router1 – Serial	Subnet D (/30)	192.168.10.113	255.255.255.252	Router-to-Router link
Router2 – Serial	Subnet D (/30)	192.168.10.114	255.255.255.252	Router-to-Router link

7. Verification and Connectivity Testing

After configuring the devices, verify the addressing and connectivity in Packet Tracer. Run the commands below on the relevant Cisco routers and PCs. Record only results obtained from the student's own lab.

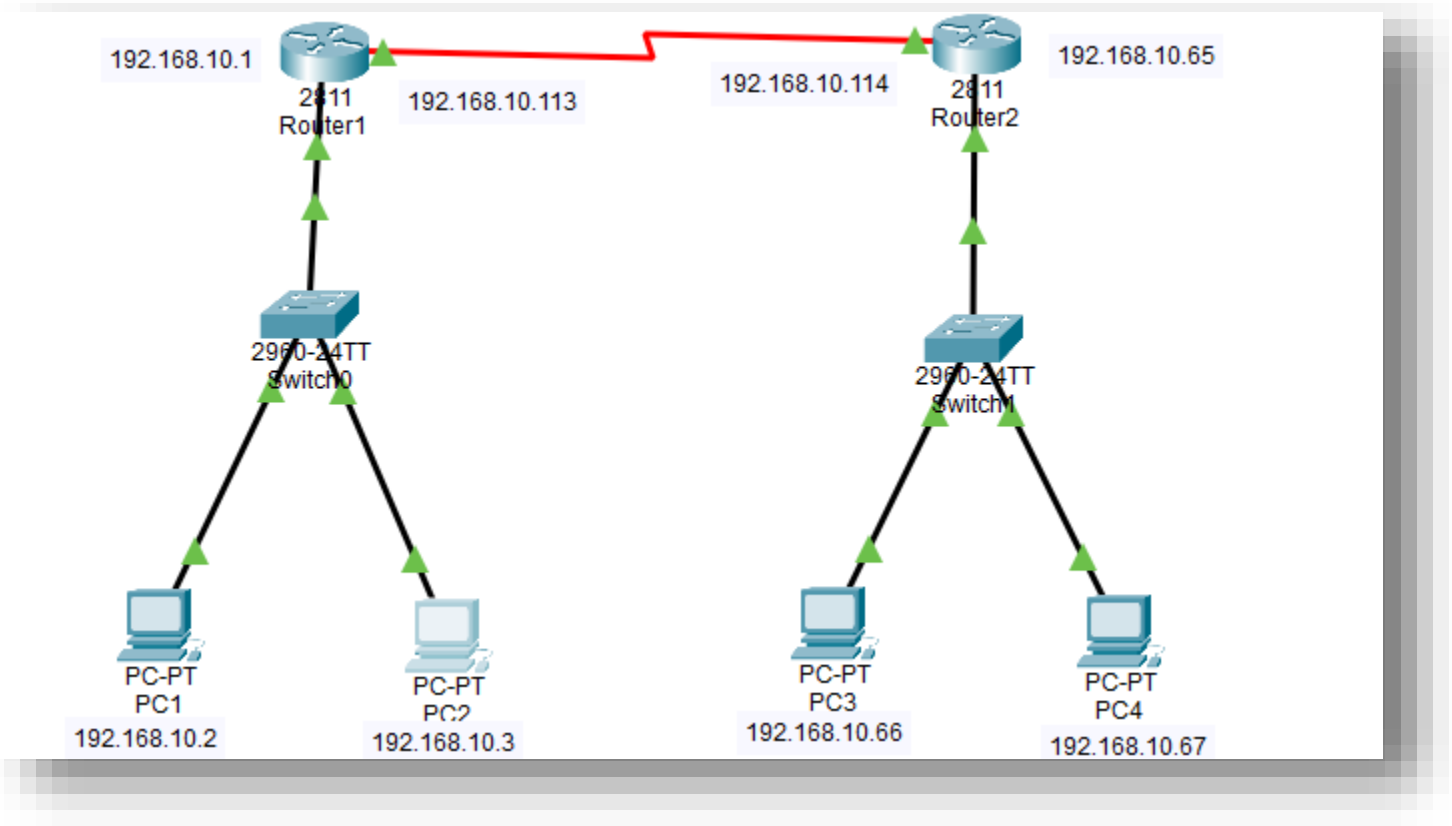
- `Router: show ip interface brief`
- `Router: show running-config`
- `Router: show ip route`
- `PC Command Prompt: ipconfig`
- `PC Command Prompt: ping <default-gateway-IP>`

- `PC Command Prompt: ping <neighbour-PC-IP>`

Test	Source	Destination	Result (Pass/Fail)	Notes
Gateway ping	PC1	192.168.10.1	Pass	Gateway reachable
Gateway ping	PC3	192.168.10.65	Pass	Gateway reachable
Neighbour ping	PC1	192.168.10.3	Pass	PC1 to PC2 successful
Inter-subnet ping (if routed)	PC4	192.168.10.67	Pass	PC1 to PC3 successful

8. Lab Screenshots — Insert Your Own Evidence

Screenshot 1 — Complete re-addressed Packet Tracer topology



Screenshot 2 — Router interface configuration / show ip interface brief

Router 1-Show ip interface brief

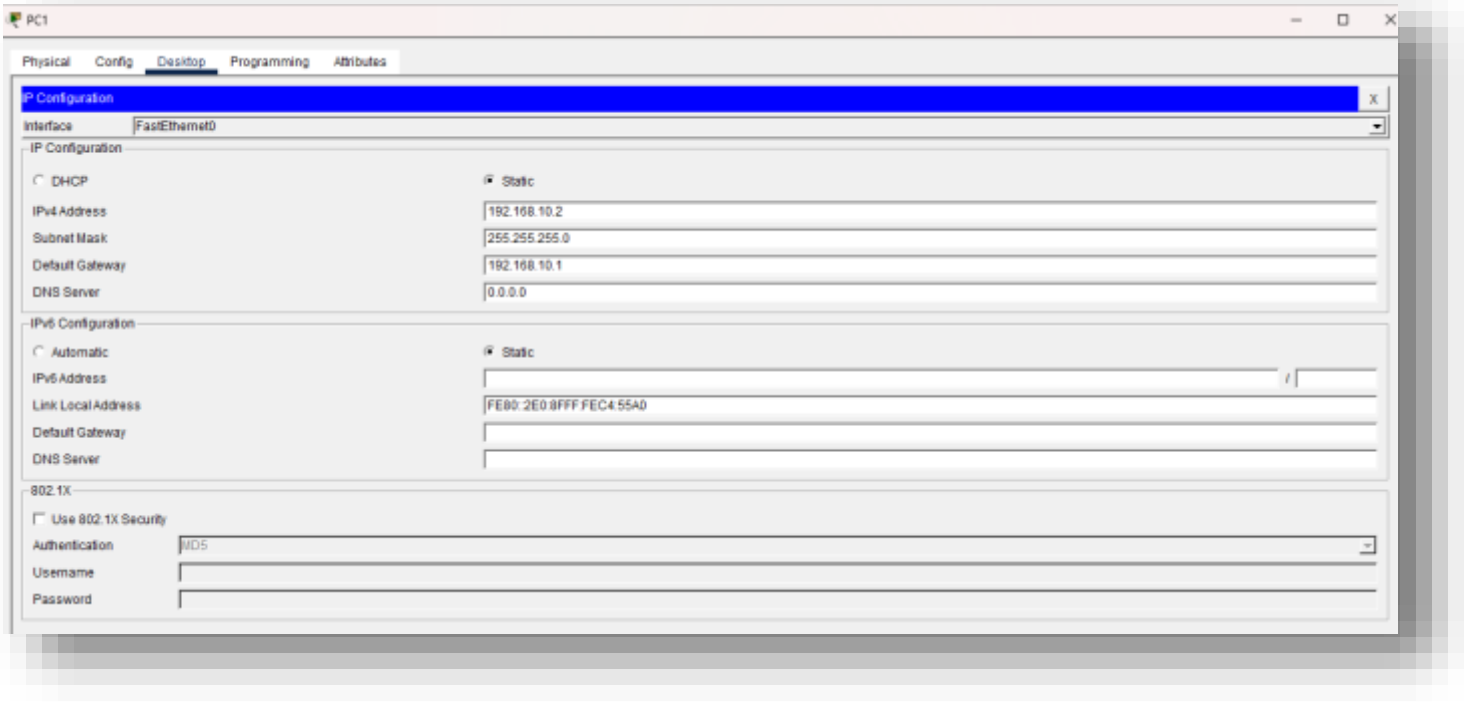
```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.65   YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            192.168.10.114  YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Router 2-Show ip interface brief

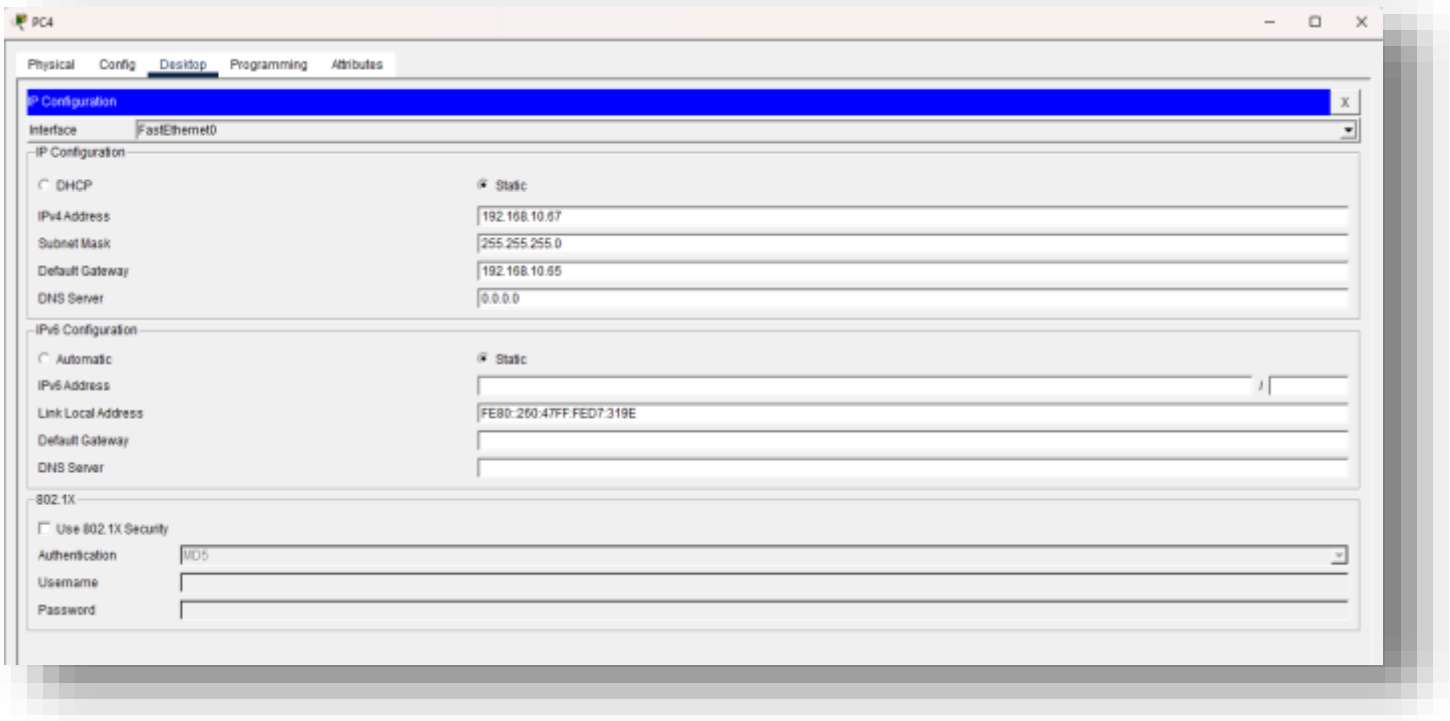
```
R1#show ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            192.168.10.113  YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Screenshot 3 — PC IP configuration

PC1 – IP configuration



PC4 – IP configuration



Screenshot 4 — Successful gateway and neighbour ping tests

Gateway Ping-Successful ping PC1 to Router 1

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Gateway Ping-Successful ping PC3 to Router 2

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time=1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Neighbour Ping- PC1 to PC2 Successful ping.

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Inter-Network Ping - PC1 to PC4 Successful ping.

```
C:\>ping 192.168.10.67

Pinging 192.168.10.67 with 32 bytes of data:

Reply from 192.168.10.67: bytes=32 time=1ms TTL=126
Reply from 192.168.10.67: bytes=32 time=4ms TTL=126
Reply from 192.168.10.67: bytes=32 time=11ms TTL=126
Reply from 192.168.10.67: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.10.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

9. Conclusion

This task demonstrates how IPv4 addresses are represented in binary, how subnet masks separate network and host portions, and how VLSM allocates address space according to host requirements. The proposed VLSM plan divides 192.168.10.0/24 into five non-overlapping subnets for 50, 25, 12, 2 and 2 hosts. The practical lab section must be completed with the actual Week 1 Task 1 device addressing and successful verification evidence.